🌐 IPv4 Addressing Part 2 (10.25.2026)

🎯 Big Picture

This topic covers how to calculate network boundaries (Network, Broadcast, Usable IPs) based on an IP address and prefix length, and how to apply these IPs to Cisco router interfaces.

🧮 The "Big 5" Calculations Cheat Sheet

To find the following, you only need to look at the **host portion** of the IP address (the bits after the prefix length):

1. **Maximum Hosts:** 2^{24} - 2 (or 2^{16} - 2, 2^{8} - 2, depending on the prefix)
    - *(Subtract 2 because the Network Address and Broadcast Address cannot be assigned to hosts).*
2. **Network Address:** Set all host bits to `0` (The very first address in the subnet).
3. **Broadcast Address:** Set all host bits to `1` (The very last address in the subnet).
4. **First Usable Address:** Network Address `+ 1`
5. **Last Usable Address:** Broadcast Address `1`

📝 Practice Examples (From Jeremy's Lab)

1. **PC1:** `43.109.23.12/8` (Class A)
    - **Network address:** 43.0.0.0
    - **Maximum hosts:** 16,777,214 *(2^{24} - 2)*
    - **Broadcast address:** 43.255.255.255
    - **First usable:** 43.0.0.1
    - **Last usable:** 43.255.255.254
2. **PC4:** `129.221.23.13/16` (Class B)
    - **Network address:** 129.221.0.0
    - **Maximum hosts:** 65,534 *(2^{16} - 2)*
    - **Broadcast address:** 129.221.255.255
    - **First usable:** 129.221.0.1
    - **Last usable:** 129.221.255.254
3. **PC8:** `209.211.3.22/24` (Class C)
    - **Network address:** 209.211.3.0
    - **Maximum hosts:** 254 *(2^{8} - 2)*
    - **Broadcast address:** 209.211.3.255
    - **First usable:** 209.211.3.1
    - **Last usable:** 209.211.3.254
4. **PC5:** `2.71.209.233/8` (Class A)
    - **Network address:** 2.0.0.0
    - **Maximum hosts:** 16,777,214 *(2^{24} - 2)*
    - **Broadcast address:** 2.255.255.255
    - **First usable:** 2.0.0.1

- **Last usable:** 2.255.255.254

5. **PC6:** `155.200.201.141/16` (Class B)
   - **Network address:** 155.200.0.0
   - **Maximum hosts:** 65,534 *(2^{16} - 2)*
   - **Broadcast address:** 155.200.255.255
   - **First usable:** 155.200.0.1
   - **Last usable:** 155.200.255.254

⚙️ Configuring IP Addresses on Cisco Devices

1. Basic Interface Configuration

You configure IPs in **Global Configuration mode -> Interface Configuration mode**.

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitethernet 0/0
Router(config-if)# ip address 10.0.0.1 255.255.255.0
Router(config-if)# no shutdown
```

- `no shutdown` is critical—interfaces are administratively down by default.
- You can use either the **dotted decimal** (`255.255.255.0`) or the **prefix length** (`/24`) when configuring the IP.

2. Adding Interface Descriptions

Good practice for documentation. Helps you remember what the interface connects to.

```
Router(config-if)# description ## Link to Switch1 ##
```

3. Verification Commands (CLI)

```
Router# show ip interface brief
Router# show interfaces
Router# show interfaces description
```

- `show ip interface brief`: Shows a quick, condensed summary of all interfaces, their assigned IP addresses, and status (up/down).
- `show interfaces` (or `show interfaces g0/0`): Shows detailed Layer 1 and Layer 2 stats (speed, duplex, MAC, input/output errors, bandwidth).

🎥 Video Reference

Free CCNA | IPv4 Addressing (Part 2) | Day 8 | CCNA 200-301 Complete Course - YouTube

- **Review of Classes/Hosts:** 1:19-3:03
- **Usable Address Calculations:** 5:39
- `show ip interface brief`: 9:34
- **Assigning IP Addresses:** 13:20
- `show interfaces`: 20:36
- **Interface Descriptions:** 22:54
- **Quiz:** 25:04-28:50